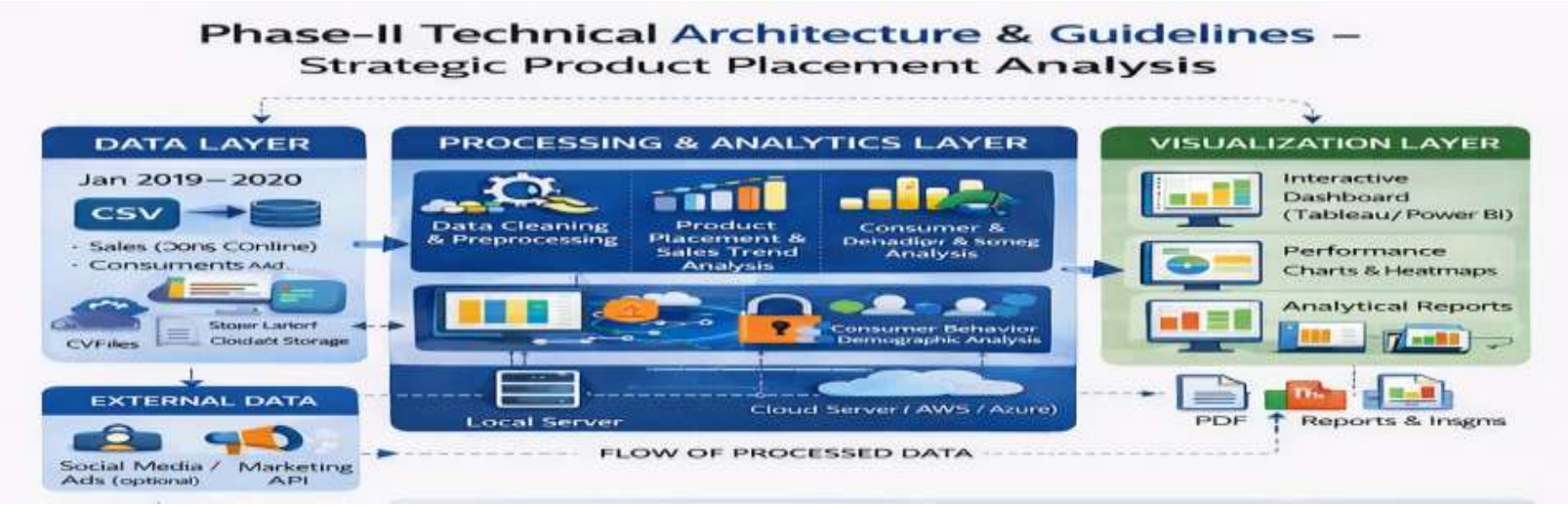
Project Design Phase-II
Data Flow Diagram & User Stories

Date	6 March 2026
Team ID	
Project Name	Strategic Product Placement Analysis
Maximum Marks	4 Marks

Data Flow Diagrams:

A Data Flow Diagram (DFD) is a traditional visual representation of the information flows within a system. A neat and clear DFD can depict the right amount of the system requirement graphically. It shows how data enters and leaves the system, what changes the information, and where data is sto



Guidelines – How the System Works

Gain actionable insights to optimize product placement strategies and drive revenue growth.

1. Data Collection

- Collect sales, placement and demographic data (2019 – 2020)

Sources:

- Sales Data (Store/Online)
- External Data:
- Social Media / · Ads

Structured Raw Dataset

2. Data Cleaning & Preprocessing

- Clean, standardize, and filter data
- Data filtering
- Data connat mistaches
- Standardize product names
- Remove duplicates.

Cleaned & Validated Dataset

3. Product Trend Analysis

- Analyze sales performance trends
- Standardize product names
- Detect outliers

Cleaned & Insghts

4. Consumer & Segment Analysis

- Segment data to find audience preferences
- Analyze customer demographics, (dge. gendet rogion)
- Track purchasing behavior & preferences.

Audience • Insights

End Result of the System:

Gain actionable **insights** to optimize product placement strategies and drive revenue growth.

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	Tools used for data analysis and visualization	Python, Pandas, NumPy, Tableau
2	Security Implementations	Secure authentication and encrypted data storage	HTTPS, Role-based Access Control
3	Scalable Architecture	System capable of handling increasing sales and consumer datasets	Cloud Deployment (AWS)
4	Availability	High availability through cloud hosting	AWS EC2 + AWS RDS
5	Performance	Efficient query processing and optimized data handling for large datasets	SQL Indexing, Data Optimization

S.No	Component	Description	Technology
1	User Interface	Interactive dashboards to visualize product placement performance and sales impact	Tableau / Power BI
2	Application Logic-1	Data Cleaning & Preprocessing of sales, product placement, and customer datasets	Python (Pandas, NumPy)
3	Application Logic-2	Product Placement & Sales Trend Analysis	Python (Matplotlib, Seaborn)
4	Application Logic-3	Consumer Behavior & Demographic Analysis	Python (Statistical Analysis, Scikit-learn)
5	Database	Structured dataset storing sales data, product placement information, and customer demographics	MySQL / PostgreSQL
6	Cloud Database	Remote database storage for scalable analytics	AWS RDS
7	File Storage	Storage of raw datasets (CSV/Excel files)	Local File System / AWS S3
8	External API	Advertising / marketing data integration	Google Analytics API / Social Media API
9	Machine Learning Model	Predict product demand and placement effectiveness	Scikit-learn / Regression Models
10	Infrastructure (Server / Cloud)	Environment to deploy analytics system and dashboards	Local Server / AWS EC2